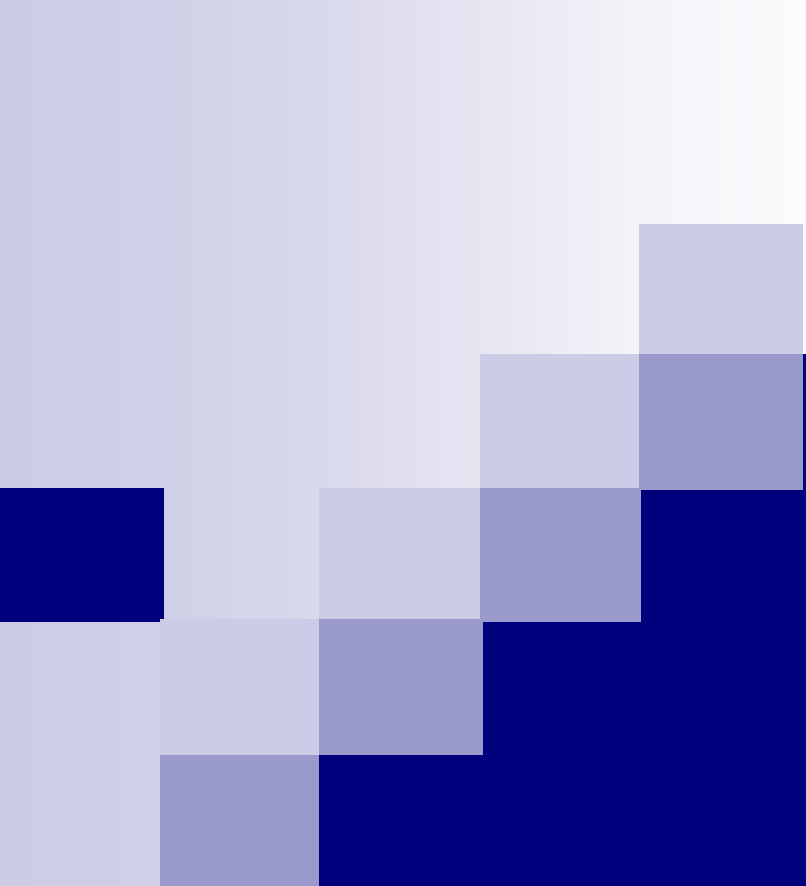


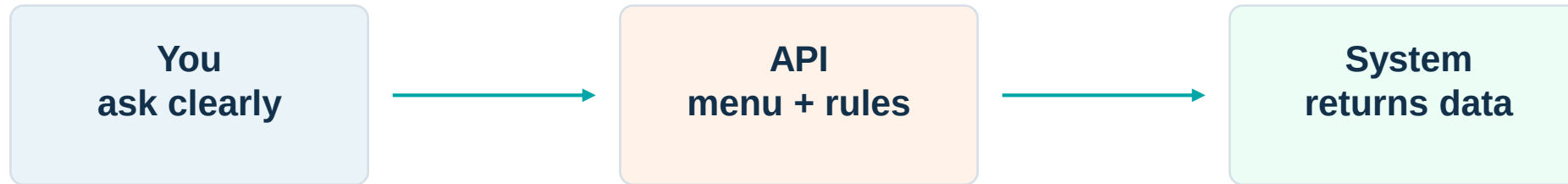


Web Analytics APIs & KPIs

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What is an API?

An API is a structured way for one system to request data or action from another system.



- Open **Google Cloud Console**.
- Sign in with your Google account.
- Create a new project.
- Go to **APIs & Services** → **Library**.
- Search for **YouTube Data API v3**.
- Click **Enable**.
- Go to **APIs & Services** → **Credentials**.
- Click **Create Credentials** → **API Key**.
- Copy the generated API key.

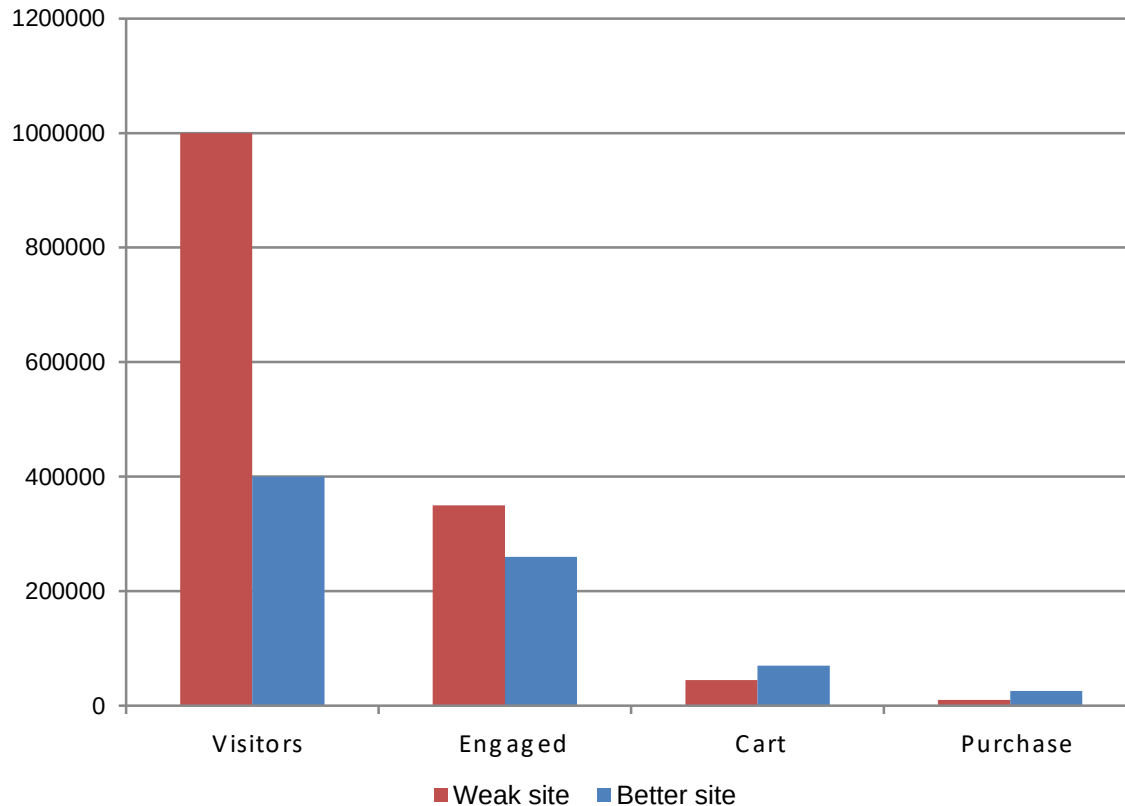
How do you know whether a website or digital platform is successful?

A website is successful when it attracts the right users, engages them, helps them complete the desired action, encourages them to return, and generates sufficient business value

Website success ≠ Traffic alone

Website success = Relevant traffic + Engagement + Conversion + Retention + Business value

If an e-commerce website gets 1 million visitors per month, can we conclude it is performing well?



More visitors do not automatically mean better business. Quality of traffic matters: source, intent, device, landing page and audience. The main job of analytics is to locate where value is lost.

Core KPI

Every metric needs a clear numerator, denominator and interpretation.

KPI	Simple formula	Interpretation
Engagement rate	Engaged sessions ÷ sessions	How many visits had meaningful activity
Bounce rate	Bounced sessions ÷ sessions	How many visits failed to engage
Pages/session	Page views ÷ sessions	How deeply users navigate
Add-to-cart rate	Add-to-cart sessions ÷ sessions	Product interest
Conversion rate	Purchases ÷ sessions	Sales effectiveness
Average order value	Revenue ÷ purchases	Basket size
Revenue/session	Revenue ÷ sessions	Monetisation efficiency
SLA breach rate	Late orders ÷ purchases	Operational reliability

Should a CEO, Marketing Manager and Website Manager monitor the same KPIs?

CEO asks:

- How much revenue has been lost?
- Has profitability declined?
- Is customer retention decreasing?
- Should more investment be made in digital channels?

Website Manager asks:

- Is the checkout page slow?
- Are mobile users facing errors?
- Is the payment gateway failing?
- At which page are users leaving?

Marketing Manager asks:

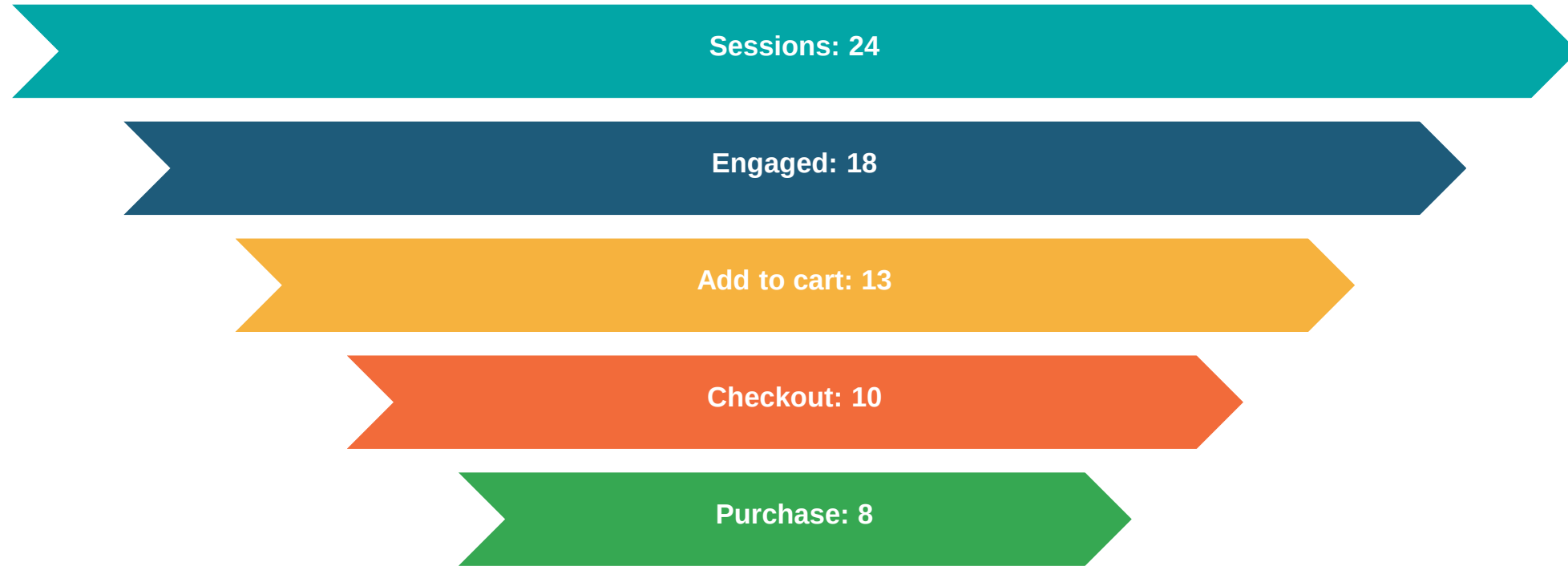
- Which campaign is underperforming?
- Has paid traffic become expensive?
- Which channel has the lowest conversion rate?
- Are we attracting the wrong audience?

CEO: Business Value

Marketing Manager: Customer Acquisition

Website Manager: Website Performance

Web analytics is a funnel, not a single number



Question

Where is the largest drop? What can the team change this week?

Case: QuickKart has traffic, but sales are weak

A quick-commerce platform must identify whether the issue is marketing, UX, pricing or operations.

QuickKart dilemma

Visitors are landing on the platform but not converting into paying customers. Management must decide where the problem lies before competitors erode market share.

Traffic is strong

Sales are weak

24

Sessions in practice dataset

- Calculate KPIs from raw session rows.
- Connect numbers to diagnosis.
- Recommend one action for each manager.

QuickKart in 24 session rows

Sheet	What to see
Session_Data	24 session-level rows with channel, device, views, duration, cart, checkout, purchase, revenue, SLA breach
Channel_Data	6 channels with impressions, clicks, spend, sessions, purchases, revenue



Unique Users
New and Returning Users
Engagement
Bounce
Average Session Duration
Pages per session

Total Sessions = 24

Unique Users = 22

New Users = 17

Returning Users = 7

Engagement Rate = Engaged Sessions ÷ Total Sessions = $18 \div 24 = 75\%$

Bounce Rate = Bounced Sessions ÷ Total Sessions = $6 \div 24 = 25\%$

Average Duration = $2,551 \div 24 = 106.3$ seconds = $106.3 \div 60 = 1.77$ minutes

Pages per Session = Total Page Views ÷ Sessions = $69 \div 24 = 2.88$ pages

24 Sessions → 18 Engaged → 13 Add to Cart → 10 Checkout → 8 Purchases



Add-to-Cart Rate

Conversion Rate

Total Revenue

Average Order Value = Revenue ÷ Purchases

Revenue per Session = Revenue ÷ Sessions

SLA Breach Rate = Late Orders ÷ Purchases



Add-to-Cart Rate = $13 \div 24 = 54.2\%$

Conversion Rate = $8 \text{ Purchases} \div 24 \text{ Sessions} = 33.3\%$

Total Revenue = ₹7,352

Average Order Value = $\text{Revenue} \div \text{Purchases} = ₹7,352 \div 8 = ₹919$

Revenue per Session = $\text{Revenue} \div \text{Sessions} = ₹7,352 \div 24 = ₹306.33$

SLA Breach Rate = $\text{Late Orders} \div \text{Purchases} = 4 \div 8 = 50\%$

CTR — Click-Through Rate

$$\text{CTR} = \text{Clicks} \div \text{Impressions} \times 100$$

CPC — Cost per Click

$$\text{CPC} = \text{Advertising Spend} \div \text{Clicks}$$

CVR — Conversion Rate

$$\text{CVR} = \text{Purchases} \div \text{Sessions} \times 100$$

CPA — Cost per Acquisition

$$\text{CPA} = \text{Advertising Spend} \div \text{Purchases}$$

ROAS — Return on Advertising Spend

$$\text{ROAS} = \text{Revenue} \div \text{Advertising Spend}$$

channel	impressions	clicks	ad_spend_i nr	sessio ns	purchas es	revenue_i nr	CTR	CPC	CVR	CPA	ROAS
Paid Search	22000	880	18000	6	0	0	4.0%	₹20	0.0%		0.00
Instagram	35000	1400	22000	5	1	549	4.0%	₹16	20.0%	₹22,000	0.02
Organic Search	12000	720	0	4	1	759	6.0%	₹0	25.0%	₹0	
Direct	0	0	0	5	4	4746			80.0%	₹0	
WhatsApp	6000	540	3000	4	3	1897	9.0%	₹6	75.0%	₹1,000	0.63
Referral	5000	250	1500	2	0	0	5.0%	₹6	0.0%		0.00

Same data, different dashboards

CEO, marketing and website manager need different KPI lenses.

Role	Primary question	KPIs to monitor	Typical action
CEO	Is digital growth profitable?	Revenue, conversion, ROAS, SLA breach	Prioritise investment and operations
Marketing manager	Which channel brings quality traffic?	CTR, CPC, CVR, CPA, ROAS	Reallocate campaign budget
Website/Product manager	Where do users struggle?	Bounce, engagement, pages/session, cart drop	Fix UX, speed, navigation, checkout
Operations manager	Can promises be fulfilled?	SLA breach, delivery time, complaints	Improve staffing, routing, dark-store capacity

A dashboard should not show everything; it should show the few indicators needed for the decision owner.

What the dataset shows

75%

Engagement rate

25%

Bounce rate

33.3%

Conversion rate

₹919

AOV

50%

SLA breach rate

0.18×

Paid ROAS

Diagnosis 1

Paid channels generate traffic but paid ROAS is weak. Marketing should not only increase traffic; it should improve traffic quality and landing-page match.

Diagnosis 2

Direct and WhatsApp convert better, suggesting trust and repeat behaviour matter. Retention and referral loops may be stronger than broad acquisition.

Diagnosis 3

Half of the purchased orders breach the 30-minute delivery promise. Operations can damage repeat visits and app ratings.

QuickKart does not have only a traffic problem; it has channel-quality, conversion-experience and delivery-reliability problems.